

Shift ciphers — revisited

We can now reinterpret shift ciphers using arithmetic in $\mathbb{Z}_n$.

Namely, by identifying $A=0, B=1, \dots, Z=25$, shift ciphers become the following cryptosystem:

- Set of plaintexts = set of ciphertexts = $\mathbb{Z}_{26}$.
- Key space = $\mathbb{Z}_{26}$.
- Encryption: for a given key $k \in \mathbb{Z}_{26}$, the corresponding encryption function is
$$e_k(x) = x + k \pmod{26}$$
- Decryption:
$$d_k(y) = (y - k) \pmod{26} =: y - k \pmod{26}$$

Claim: This is a cryptosystem, i.e. $d_k(e_k(x)) = x$ for all $x, k \in \mathbb{Z}_{26}$.

Pf: It's enough to show that $d_k(e_k(x)) \equiv x \pmod{26}$.

$$\begin{aligned}
 \text{Indeed, } d_k(e_k(x)) &= d_k((x+k) \pmod{26}) \\
 &= ((x+k) \pmod{26} - k) \pmod{26} \\
 &\equiv ((x+k) \pmod{26}) - k \equiv (x+k) - k \\
 &\equiv x \pmod{26}. \quad //
 \end{aligned}$$

Of course, there is nothing special about "26" and this idea works over arbitrary finite alphabets! 21

As indicated above, division in $\mathbb{Z}_n$ is not merely a shadow of division in $\mathbb{Z}$.

Defn An inverse of $a \in \mathbb{Z}$ modulo n (or a modular inverse of a modulo n) is any integer $b \in \mathbb{Z}$ with

$$ab \equiv 1 \pmod{n}.$$

Ex: • Since $3 \cdot 11 = 33 \equiv 1 \pmod{32}$, 3 is an inverse of 11 (and vice versa) modulo 32.

Note that e.g. $35 = 3 + 32$ is another inverse of 11 mod 32.

• 2 does not admit an inverse modulo 4. Why?

Fact • An integer $a \in \mathbb{Z}$ admits an inverse modulo n iff $\gcd(a, n) = 1$.
• If b, b' are inverses of a modulo n , then $b \equiv b' \pmod{n}$.

(Recall: the greatest common divisor $\gcd(a, b)$ of a and b (not both zero) is the largest $c > 0$ with $c \mid a$ and $c \mid b$.)

Recall that given a, n the Extended Euclidean Algorithm ²²

computes $x, y \in \mathbb{Z}$ with $ax + ny = \gcd(a, n)$.

Hence, if a admits an inverse modulo n , then $\gcd(a, n) = 1$ and x is an inverse of a modulo n .

Later on, we'll often require "random elements of $\mathbb{Z}_n$ that admit a modular inverse" for cryptographic applications. How can we find those?

Defn Euler's totient function:

$\varphi(n) :=$ no. of elements of $\mathbb{Z}_n^\times := \{a \in \mathbb{Z}_n : \gcd(a, n) = 1\}$

("unit group" of $\mathbb{Z}_n$)

Facts • $1 \leq \varphi(n) \leq n$ (obvious)

• $\limsup_{n \rightarrow \infty} \frac{\varphi(n)}{n} = 1$ (i.e. $\varphi(n) \approx n$ infinitely often; e.g. for n prime)

• $\frac{\varphi(n)}{n^{1-\delta}} \rightarrow \infty$ as $n \rightarrow \infty$, for each $\delta > 0$

(That is, $\varphi(n)$ doesn't grow much slower than n .
→ Hardy & Wright.)

Affine cipher (1-dim. case)

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Using modular inverses, we can build a cryptosystem that generalises shift cipher:

- Set of plaintexts = set of ciphertexts = $\mathbb{Z}_n$
- Keys are pairs $(a, b) \in \mathbb{Z}_n^\times \times \mathbb{Z}_n$ ($= \text{AGL}_1(\mathbb{Z}_n)$ for group theorists)

Hence: $\# \text{Keys} = \varphi(n)n$

- Encryption: $e_{(a,b)}(x) := a \cdot_n x +_n b = (ax+b) \bmod n$
Key $\nearrow$ message from $\mathbb{Z}_n$

- Decryption:
let $\bar{a} \in \mathbb{Z}_n$ be the unique (!) inverse of a modulo n .

Then $d_{(a,b)}(x) := \bar{a} \cdot_n (x -_n b) = \bar{a}(x-b) \bmod n$.

This is indeed a cryptosystem since

$$\begin{aligned} d_{(a,b)}(e_{(a,b)}(x)) &= \bar{a} \cdot ((a \cdot x + b) - b) = \bar{a} \cdot (a \cdot x) \\ &\quad \text{all operations mod } n! \\ &= x \quad \text{for all } x \in \mathbb{Z}_n \end{aligned}$$

Notes • For $a=1$ (hence $\bar{a}=1$), we recover shift ciphers. 24

• Affine ciphers are vulnerable to frequency analysis and hence insecure

Ex: $n=26$, $a=3$, $b=5$ (NOTE: $\gcd(3, 2 \cdot 13) = 1$)

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Plain text: TEXT $\rightsquigarrow$ KRWK
19 4 23 19

$$T \leftrightarrow 19 \rightsquigarrow (3 \cdot 19 + 5) \bmod 26 = 62 \bmod 26 = 10 \leftrightarrow K$$

$$E \leftrightarrow 4 \rightsquigarrow (3 \cdot 4 + 5) \bmod 26 = 17 \leftrightarrow R$$

$$X \leftrightarrow 23 \rightsquigarrow (3 \cdot 23 + 5) \bmod 26 = 22 \leftrightarrow W$$

Next: Higher-dimensional affine ciphers

Reminder: Matrix multiplication let $M_{d \times e}(R)$ be the set of $d \times e$ matrices with entries in R (e.g. $\mathbb{Z}, \mathbb{Q}, \mathbb{C}, \mathbb{Z}_n$).
↑ rows ↑ columns

When R is a ring (so that $+$ and $\cdot$ are defined in R , e.g. $R = \mathbb{Z}$ or $R = \mathbb{Z}_n$), the product AB of $A \in M_{d \times e}(R)$ and $B \in M_{e \times f}(R)$ is the $d \times f$ matrix with (i, j) entry $\sum_{k=1}^e a_{ik} b_{kj}$ (where $A = [a_{ij}]$ and $B = [b_{ij}]$).